

PLANNING, SECURITY, AND YOUR FIRST REPOSITORY ENTRY

Submission Form

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Course/Section: IT PROF ELEC 4 - 80107	Date: SEPTEMBER 2, 2026

Chosen App or Organization (semester project): GCASH

Part 1. Planning Question Evaluation

New request being evaluated (one sentence): Develop a real-time fraud alert system for GCash that detects unusual transactions and immediately notifies customers of potentially suspicious activities.

1. Does this solve a real problem?

Yes. GCash customers may experience suspicious transactions or unauthorized account activity. A real-time fraud alert system can help customers become aware of unusual transactions quickly so they can take appropriate action, such as checking their account or contacting GCash support.

2. Do we already have something that can do this?

Yes. GCash already has security features and transaction monitoring that can help protect customers from suspicious activities. However, the proposed system can improve the process by providing faster and more specific alerts when a transaction appears unusual based on factors such as transaction amount, frequency, location, or device activity.

3. Does this support the organization's actual goals?

Yes. A fraud alert system supports GCash's goal of providing secure and reliable financial services to its customers. It can help improve customer awareness of suspicious activities, strengthen account security, and reduce the possible impact of unauthorized transactions.

4. Is it secure and cost-effective?

It can be secure and cost-effective if the system uses appropriate security controls. The system should protect customer and transaction data through encryption, authentication, access controls, secure APIs, and monitoring. Using GCash's existing transaction and authentication infrastructure can also reduce the need to build completely separate systems.

Where would this rank against other approved work? (value, risk, dependency, or effort, reasoned in words, not calculated)

This would be a high-priority security-related project because it directly concerns customer transactions and account protection. However, it would require careful integration with existing GCash transaction systems and reliable detection rules to avoid unnecessary or incorrect alerts.

Part 2. Security Pass Across the Four Domains

Domain	Security Concern	Matching Control
Business	Unauthorized employees could access fraud-monitoring functions or change detection rules without approval.	Use role-based access control, approval procedures, separation of duties, and audit logs for administrative activities.
Data	Sensitive customer and transaction information used for fraud detection could be exposed, stolen, or misused.	Use encryption, data minimization, access restrictions, secure storage, and appropriate data-retention policies.
Application	Attackers could manipulate the fraud detection system or exploit vulnerabilities to prevent alerts or generate false alerts.	Use secure authentication, authorization, input validation, API security, logging, monitoring, and regular security testing.
Technology	Servers, databases, APIs, or other infrastructure supporting fraud detection could become vulnerable because of outdated software or insecure configurations.	Apply security patches, secure configurations, vulnerability scanning, continuous monitoring, backups, and access restrictions.

One weak-link risk (how a weak point in one part could put a stronger part at risk):

If the fraud detection application has weak authentication or access controls, an unauthorized person could modify fraud detection rules or disable alerts. Even if the database and servers are strongly protected, a weakness in the application could allow attackers to interfere with the fraud detection process.

Part 3. Your First Repository Entry

Component being documented (application, data store, or technology platform):

GCash Fraud Detection and Alert System (Application)

Field	Your Entry
What it does	Monitors GCash transactions for unusual or potentially suspicious activity using factors such as transaction amount, frequency, location, or device activity. When suspicious activity is detected, the system sends real-time alerts to customers and authorized personnel for appropriate investigation or action.
Which department or user group uses it	GCash customers receive alerts about potentially suspicious transactions, while authorized fraud-monitoring, cybersecurity, customer support, and IT personnel may use the system to monitor, investigate, and respond to suspicious activities.
What it connects to	It connects to GCash's transaction processing system, customer and account database, authentication system, fraud or risk detection services, notification service, and approved APIs.
Who is responsible for it	GCash's IT and cybersecurity teams are responsible for the technical operation and security of the system, while authorized fraud and security personnel are responsible for monitoring and responding to suspicious activities.
Whether it is still needed, and why	Yes. It is needed because detecting unusual transactions and providing real-time fraud alerts can help customers become aware of potentially suspicious activities quickly. It also supports account protection and the security of digital financial services.